Ao Guo

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Research Interests

Black Hole Superradiance, Gravitational-Wave Astrophysics and Numerical Relativity

Research Experiences

- **Projects related to superradiance**

  Advisors: Prof. Jun Zhang (International Centre for Theoretical Physics Asia-Pacific),
  Collaborator: Prof. Huan Yang (Tsinghua University)

  Develop analytical models of ultralight-boson clouds around rotating black holes and study their evolution and observational signatures in compact binaries.

  - **The common envelope phase preserves clouds from depletion[1](2024)**

    Showed that common-envelope evolution can allow binaries to rapidly cross key depletion channels, preserving clouds into the LISA band.

  - **Mass transfer of superradiant clouds[1](2024)**

    Derived an analytical description of resonant cloud transfer to a binary companion, including transfer efficiency and resonant radii. Demonstrated a pathway for non-superradiant objects, such as neutron stars, to host boson clouds and enable multimessenger searches for ultralight bosons.

  - **Formation and evolution of the common envelope[2](2025)**

    Developed a semi-analytical gravitational-molecule framework for common-envelope formation and evolution in black-hole binaries with general mass ratios. Found that back-reaction from the evolving envelope can excite orbital eccentricity to O(0.1) in the LIGO–Virgo–KAGRA band.

- **Conversion from gravitons to photons[3](2022)**

  - Studied high-frequency gravitational-wave conversion into photons in the dipole magnetospheres of neutron stars. Identified a potential multimessenger probe of high-frequency gravitational waves and dark matter.

- **Response of regular black holes to tidal perturbations(Undergraduate Thesis)**

  Analyzed the response of regular black holes(Hayward, Bardeen and Simpson-Visser metrics) to scalar and vector tidal perturbations

Publications

[1] **A. G.**, Jun Zhang, and Huan Yang. "Superradiant clouds may be relevant for close compact object binaries". *Phys. Rev. D* 110.2 (2024), p. 023022. DOI: 10.1103/PhysRevD.110.023022. arXiv: 2401.15003 [gr-qc].

[2] **A. G.**, Qi-Yan Zhang, Huan Yang, and Jun Zhang. "Common envelope evolution of ultralight boson clouds". *Phys. Rev. D* 113.4 (2026), p. 043018. DOI: 10.1103/g79c-mqgx. arXiv: 2508.18738 [gr-qc].

[3] Chao-Jun Feng, **A. G.**, Zhong-Ming Xie, and Miao Li. "Photons generated by gravitational waves in the near-zone of a neutron star". *Nucl. Phys. B* 985 (2022), p. 116029. DOI: 10.1016/j.nuclphysb.2022.116029. arXiv: 2207.14517 [gr-qc].

Conferences

- New horizons for Psi school & workshop, Instituto Superior Técnico, Portugal
- CTPU-CGA 2025 Workshop on Quasi Normal Mode and Black Hole Perturbation, Institute for Basic Science, Korea(My presentation)

Education

Sept 2026 - Present	PhD, Mathematics Queen Mary University of London, England
Sept 2023 - Jun 2026	Master, Theoretical Physics International Centre for Theoretical Physics Asia-Pacific(ICTP-AP), University of Chinese Academy of Sciences, China
Sept 2019 - Jun 2023	Bachelor, Physics Southern University of Science and Technology, China

Awards & Honors

2025 **National Scholarship**(The sole recipient of the college as a master student)
2025 Merit Student Award of University of Chinese Academy of Sciences
2023 **Outstanding Dissertation** of Southern University of Science and Technology
2022 National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling
2021 National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling

Coding Skills

Mathematica Python Matlab Java *C*

Leadership & Service

- Served as a volunteer for the Symposium on the History of Modern Physics in 2024
- Served as the president of the red cross society and the piano club on campus
- Performed a piano solo at the Physics Night event of 2024 International Congress of Basic Science